



Developing a Process Simulation Model for Lipid Nano-particle Manufacturing



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Motivation & Background

mRNA vaccines (Pfizer-BioNTech, Moderna) saved millions of lives. Their secret ingredient: Lipid Nanoparticles (LNPs) - tiny spheres that protect and deliver mRNA into human cells.

Challenges

LNP production involves dozens of interlinked variables-flow rates, lipid ratios, pH, filtration pressures - making it extremely complex to design and scale up. So, we need an easy-to-use tool to calculate all process quantities.

Our Solution

We built a process simulation model that handles all these calculations for you.

Process Analytical Technology (PAT)

The FDA defines PAT as a system to analyze, monitor, and control pharmaceutical manufacturing processes in real time. We applied PAT by designing a P&ID with real control loops not just offline testing.

FOUR MEASUREMENT LEVELS:

INLINE	Real-time, no sample removal. Eg: DLS, Zeta potential ,pH sensor, Conductivity, Flow meter, Temperature probes, Pressure probes.
ONLINE	Near real-time, automated sample removal Eg: HPLC, Raman spectroscopy, LC-MS, UV diode array.
ATLINE	Manual sample removal near production line. Eg: DLS, Zeta potential, NMR.
OFFLINE	Sample removed to separate lab. Eg: Cryo-TEM.

Focus on identifying two basis:

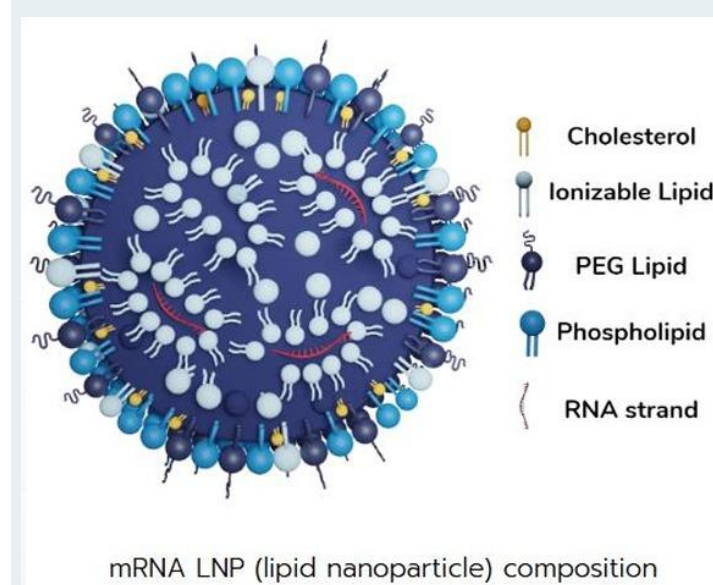
- 1) CQAs (Critical Quality Attributes). Eg: Size <100 nm, PDI <0.3.
- 2) CPPs (Critical Process Parameters). Eg: Flow, Temperature, Pressure etc.

Key Software Feature

- **Three Calculation Modes:** FORWARD (feeds → product), BACKWARD (desired product → required feeds), or BOTH simultaneously.
- **DOF Analysis:** Automatically checks the Degrees of Freedom. Alerts user if the system is under- or over-specified before solving.
- **Self-Consistency Check:** If concentration AND mass flow are both given for the same component, the software verifies they are consistent with the flow rate.
- **Warnings & Validation:** Flags physically impossible values (e.g. negative flows), out-of-range conditions, and concentration/mass mismatches with severity levels.
- **Material Balance Verification:** Checks that total mass IN equals total mass OUT for every component at TFF level. Reports PASS/FAIL per component.
- **Knowability Scoring:** In over-determined cases, automatically selects which variable to treat as redundant, based on how directly measurable each is.
- **Run History & Reports:** Every run is saved in a timestamped folder: output report, diagnostics file, and a copy of the input used.

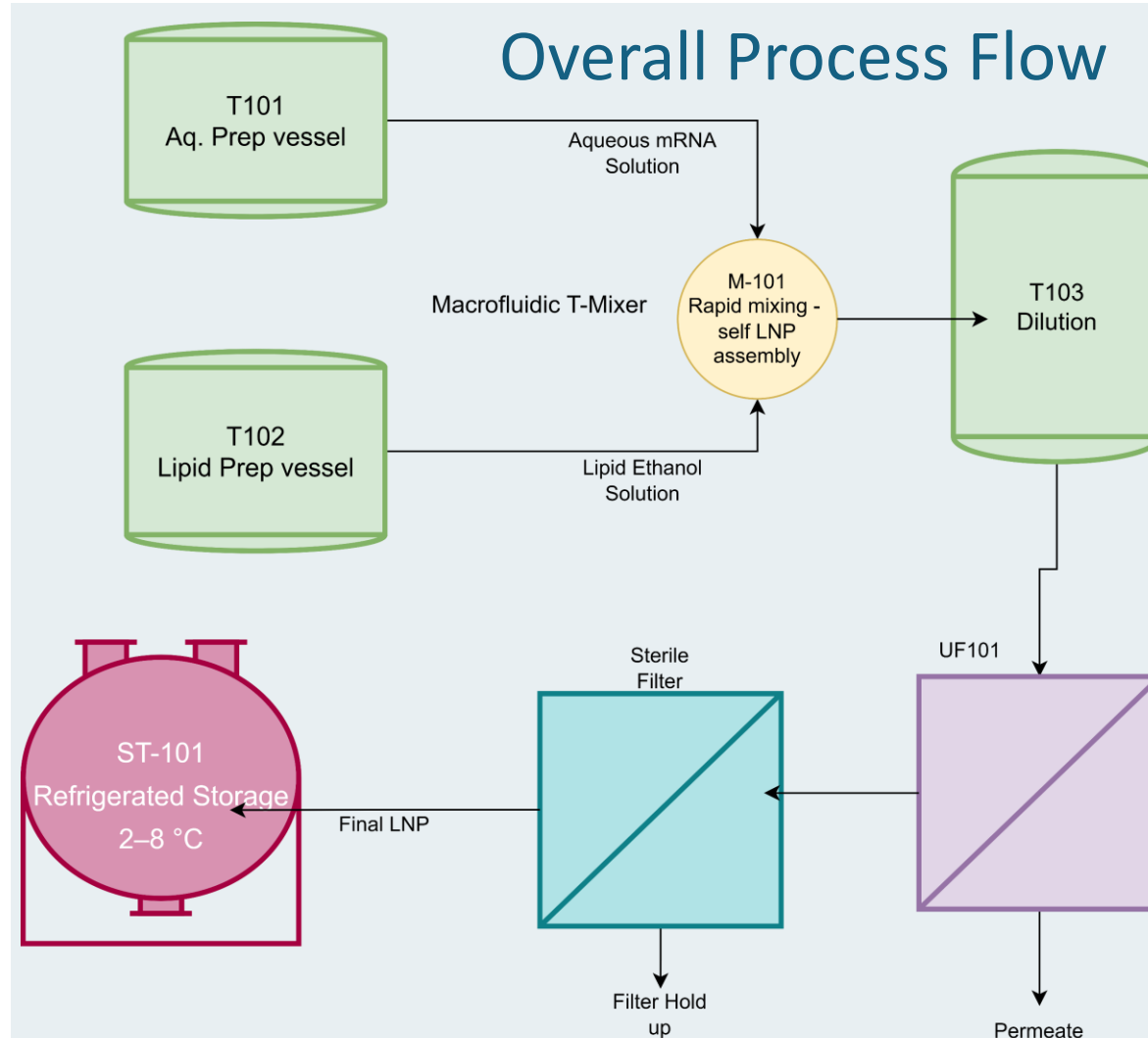
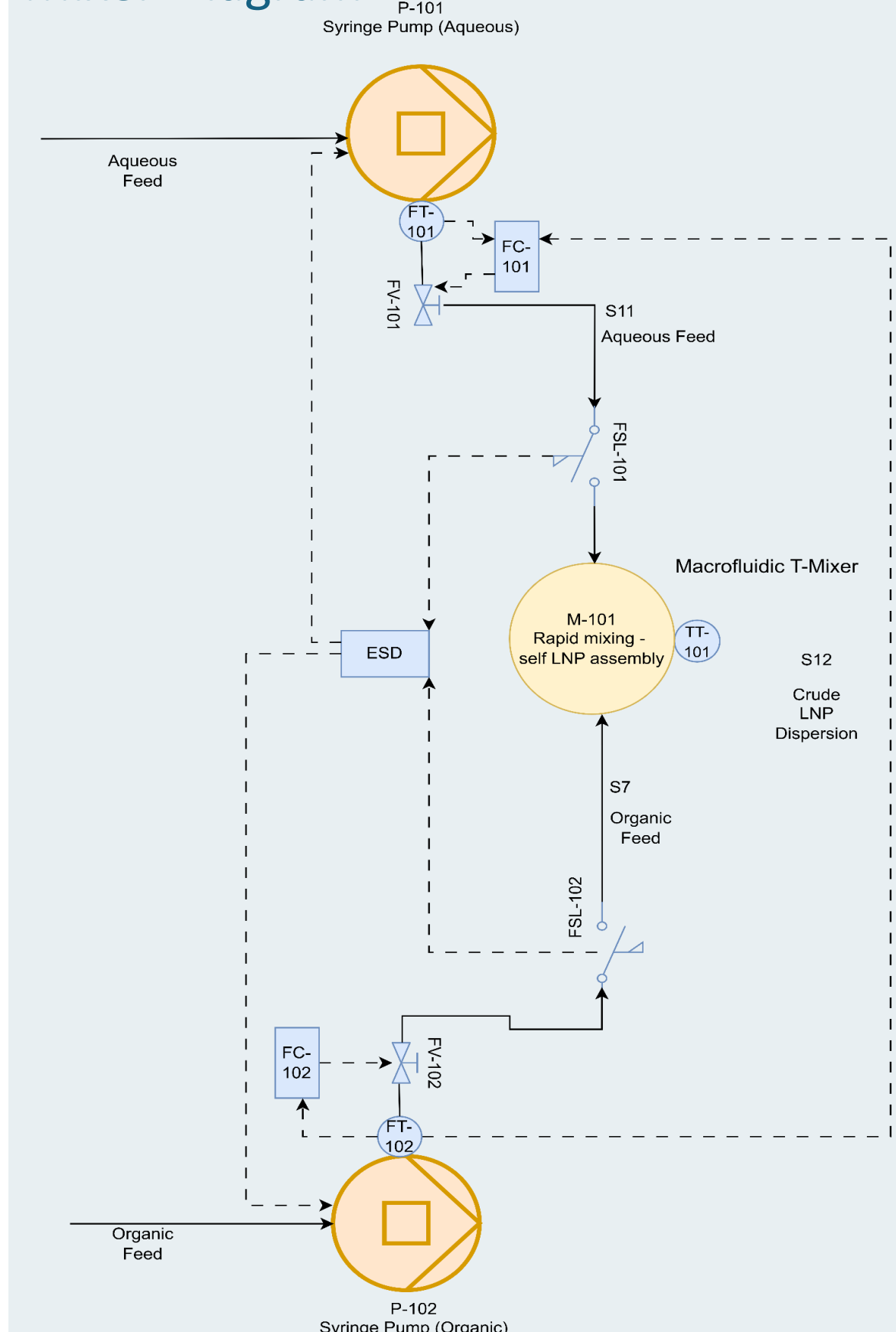
Lipid Nano-particles

- LNPs are tiny, spherical particles made of lipids of size - (10nm - 100nm).
- They encapsulate and protect fragile molecules like mRNA to reach and enter in cells efficiently.

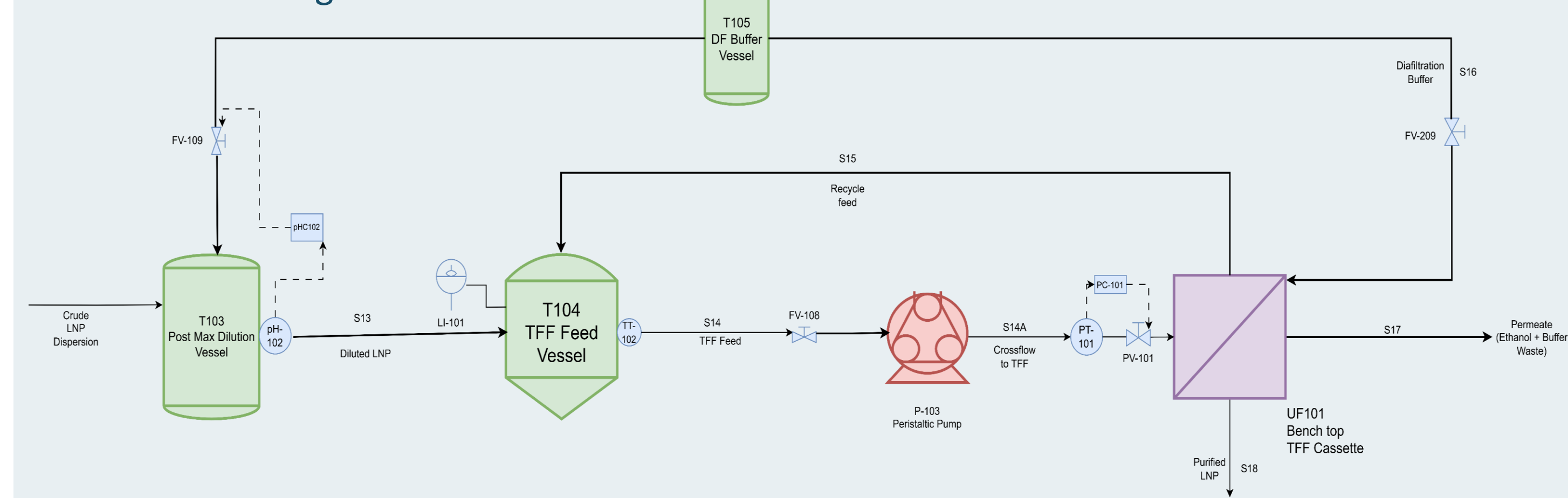


- **Ionizable lipid:** binds RNA and enables endosomal escape.
- **Phospholipid:** provides structural stability.
- **Cholesterol:** improve membrane rigidity and integrity.
- **PEG-lipid:** prevents aggregation and improves circulation time.

Mixer Diagram



TFF & Dilution Diagram



Simulation Logic — Over-determined Case (DOF < 0)

1. User gives Input
2. Model read input & resolve_frr() derive Aqueous, Organic or FRR from any two of the three
3. DOF Analysis: $DOF = 2 - n_{\text{known}}$ **ERROR:** Under-determined Need more known values
4. $DOF < 0 \rightarrow$ Knowability Scoring Rank known variables by how directly measurable (pump settings=0.95 highest; derived flows=0.45 lowest)
5. build_system() Build 7x9 matrix A from EQ1 EQ7 (Aq+Org=Mix Cryo-dead=Prod)
6. Redundancy Verification Compare: solver_value vs user_value for each dropped var. PASS if $|diff| < 1e-6$
7. verify_balance() Check all 9 equations hold (EQ1–EQ7, FRR check, Overall balance) PASS/FAIL per equation
8. identify_mode() $\rightarrow$ FORWARD / BACKWARD / GENERAL write_report() + write_diagnostics() $\rightarrow$ timestamped folder
9. OUTPUT: All stream flows, Concentrations Warnings, Balance PASS/FAIL, Saved reports

LNP Component Library (150+ Compounds)

We catalogued every compound used in LNP manufacturing from published patents and scientific literature, with molecular weights and clinical annotations:

14 Ionizable Lipids SM-102 · ALC-0315 · DLin-MC3-DMA · C12-200 · cKK-E12...	19 Helper Lipids DSPC · DOPE · DPPC · POPC · Sphingomyelin · SOPC...
47+ Sterols Cholesterol + 46 derivatives, oxysterols & phytosterols	18 PEG-Lipids PEG2000-DMG · ALC-0159 · DSPE-PEG1000/3000/5000...
15 Cryoprotectants Sucrose · Trehalose · Mannitol · Raffinose · Dextran...	14 Buffer & Salt Systems Citric acid · Histidine (Pfizer) · Tris (Moderna) · HEPES...
8 Antioxidants & Chelators Tocopherol · EDTA · Methionine · Glutathione · Ascorbic acid	14 Surfactants & Stabilisers Polysorbate 20/80 · Poloxamer 188 · Glycerol · PVA...

Future Work

- **Particle Size Prediction:** Add empirical correlations: FRR + total flow rate $\rightarrow$ predicted LNP size and PDI.
- **Experimental Validation:** Verify all model predictions against physical LNP synthesis runs in the laboratory.
- **Digital Twin:** Connect PAT sensor feeds to simulation in real-time - close the loop between model and process.
- **DoE Optimisation Module:** Automated formulation screening $\rightarrow$ find optimal lipid ratios for target CQAs.
- **Scale-Up Calculator:** Extend from lab syringe pumps to industrial IJM (KNAUER) and NxGen (Precision Nanosystems) systems.

Acknowledgement

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